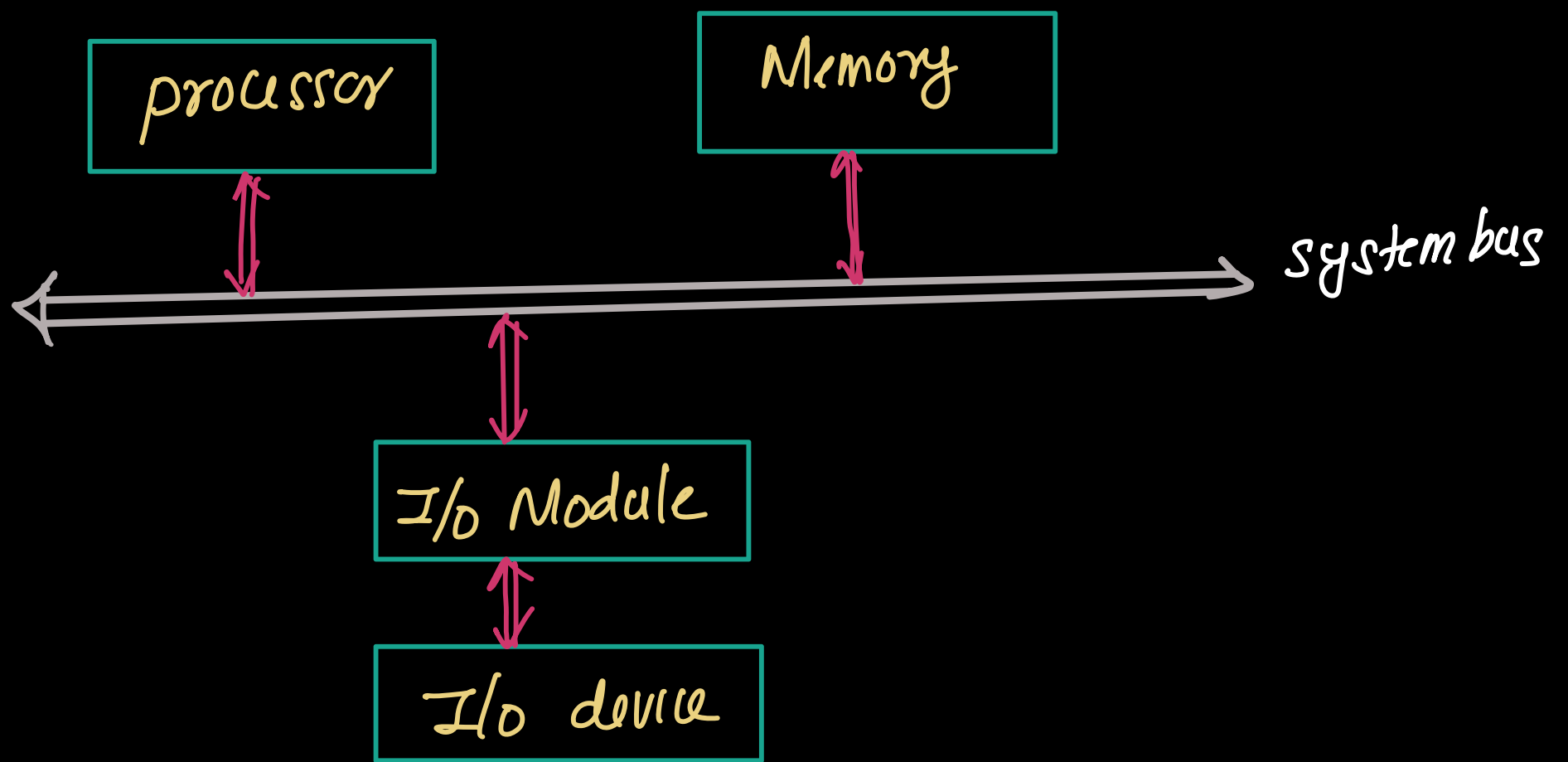
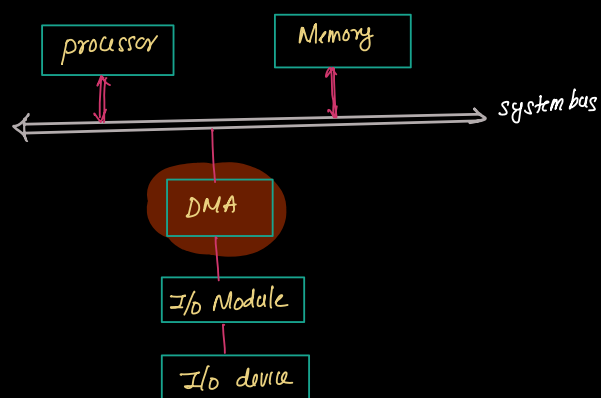


Direct Memory Access (Note)

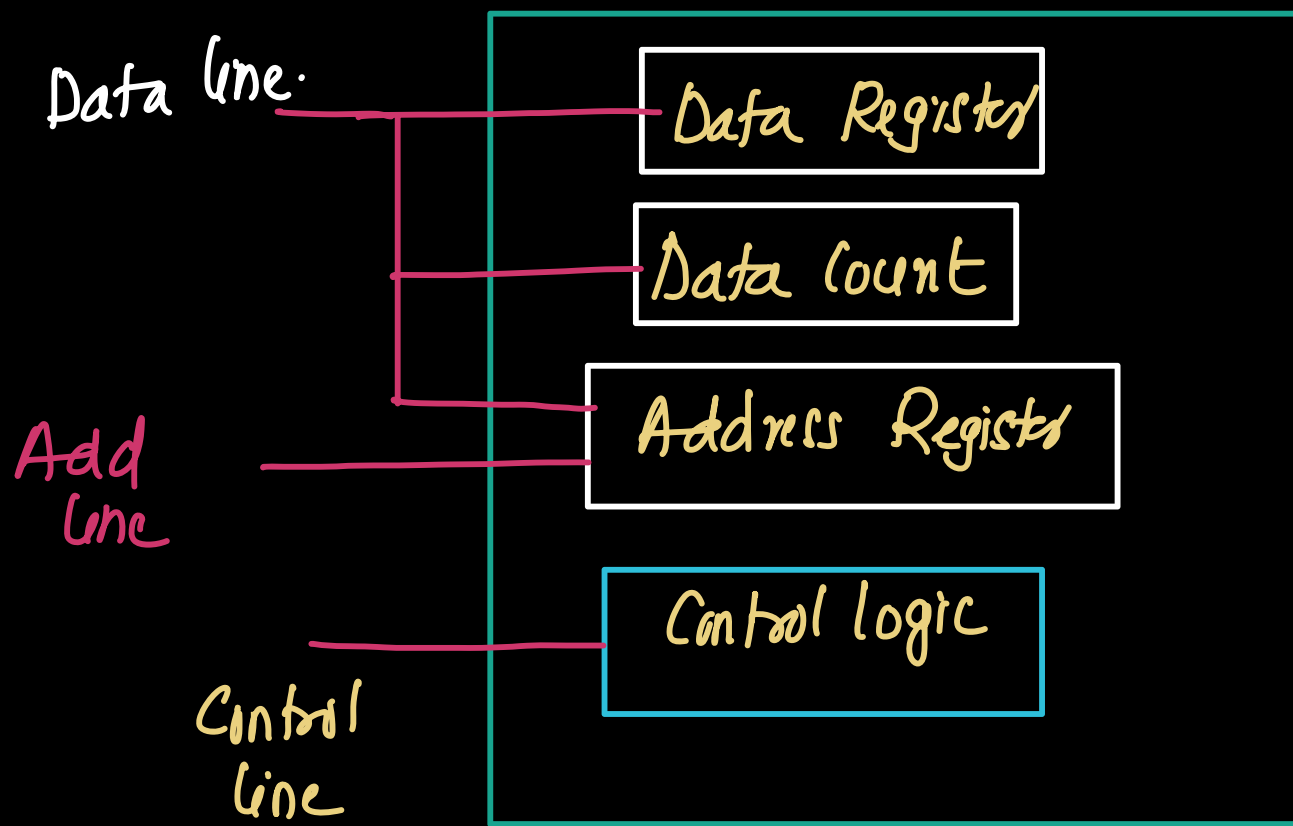


- Consider an I/O operation in which block of data must be read from the I/O device and store the data in memory.
- In a computer system, processor, memory, I/O device, I/O module are connected through the system bus.

- If programmed I/O is used for this I/O operation, processor is completely dedicated to managing the data transfer. processor must repeatedly check the status of I/O device and transfer each word of data b/w device and memory. During this time processor cannot perform other useful task. therefore processor is not used efficiently.
- In interrupt driven I/O, the processor issues an I/O command to I/O module. and processor performs other task. When the device becomes ready, it generates an interrupt and processor performs the data transfer. Although this method improves processor utilization, processor must still participate in each data transfer b/w I/O module and memory.
- When a large block of data must be transferred, both programmed I/O and interrupt driven I/O are inefficient. Therefore a technique is required that allows the I/O operation (data transfer b/w I/O device and memory) to occur without the continuous involvement of the processor. This technique is called Direct Memory Access (DMA).
- DMA allows data to be transferred directly b/w I/O devices and memory without continuous involvement of the processor.
- In this method special hardware unit called DMA module (DMA Controller) is connected to the system bus.



Block diagram of DMA



DMA function

(W)

— Consider an i/o operation in which block of data must be read from an i/o device and stored in memory.

— Step-1

When processor wants to read a block of data from an i/o device, it issues a command to the DMA module. The processor send following information to the DMA

① Address of I/O device/i/o module

The processor send the add of I/O device so that DMA controller knows which device will participate in the data transfer.

(2) The processor specify the operation need to perform (Read / write) using control signal through control lines of processor. This control signal is interpreted by control logic inside the DMA.

(3) processor send start Add of memory location where the data will be stored. This Add is stored in the Add register of DMA control.

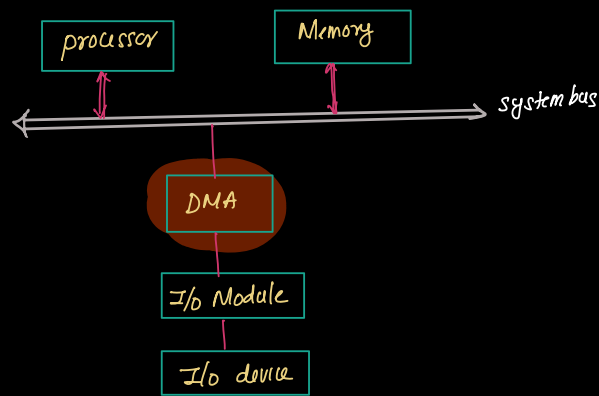
(4) processor send the number of words (data) to be transferred to Count Register inside the DMA.

Step-2

→ After providing all the information, processor continue executing other instruction. Thus DMA coordinates the data transfer operation.

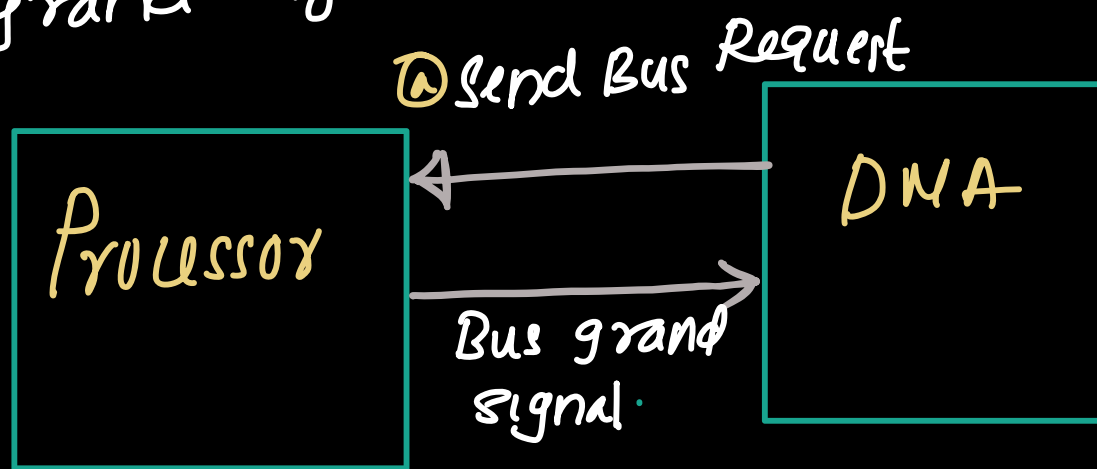
Step-3

— In a computer system processor and the DMA share the same system bus to access the memory. (10)



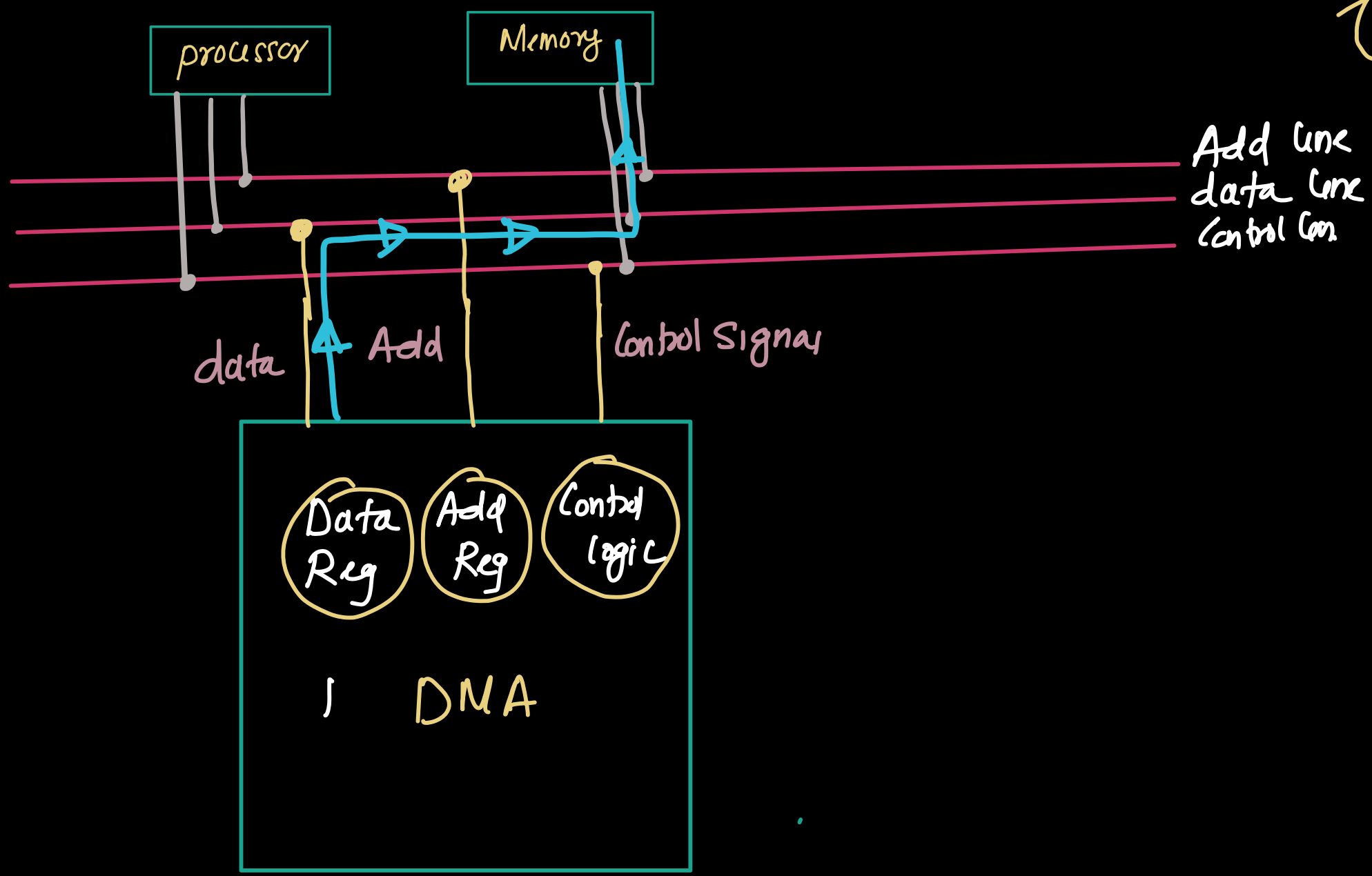
— Since only one component can use the bus at a time.

→ To transfer data btw I/O device and memory, DMA send a Bus Request signal to the processor. Processor completes the current bus operation and then grants control of system bus to DMA by sending Bus grant signal.



Step-4 — DMA transfer the one block of data directly btw I/O module and memory.

②



Step 5: After transferring one word, DMA Controller releases the bus, allowing the processor to continue its execution.

this technique is called bus stealing, because the DMA controller steals one bus cycle from the processor to perform the task.

eg: instruction execution includes following steps

- ① Fetch — Fetch instruction
- ② Decode — Decode instruction
- ③ Fetch operands —
- ④ execution → perform the task
- ⑤ Memory access →

In the Fetch, Fetch operand, memory access stages use s/m bus and decode, execution not using s/m

Fetch	Decode	Fetch operand	Execute	Memory access
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Let say processor finish decoding, Next step is Fetch operands. Before starting Fetch operand processor need s/m bus to access the memory. At this moment DMA request has, processor pauses temporarily. DMA transfer one word of the data. The bus is returned to the processor. Then the processor continue fetch operand.

Step-6 — DMA continues transferring data one word at a time until the data count register become zero. this indicate that entire block has been transferred.

Step-7 — After the transfer is complete, DMA send ^(N) an interrupt signal to the processor to indicate that DMA operation finished

Summary:

- ① processor send information to DMA (send command)
- ② processor continue execution of other instruction
- ③ DMA request system BUS (sending Bus request signal)
- ④ processor complete current bus ^{operation} and send bus grant signal
- ⑤ DMA transfer the data (stealing bus cycle) one word at a time
- ⑥ DMA complete the data transfer and send interrupt signal to the processor.

Different DMA Configuration.

②

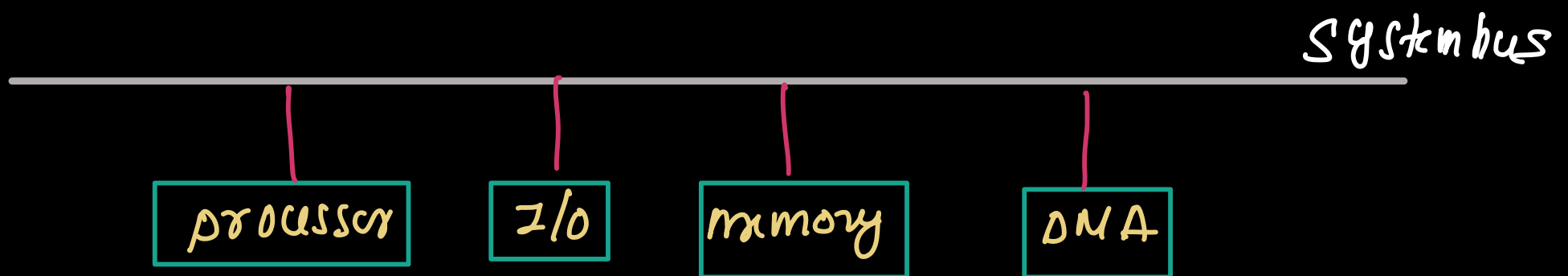
— this represents how components processor, Memory, I/O devices, DMA are connected.

— 3 Configurations are possible.

- ① Single bus detached DMA
- ② Single bus integrated DMA - I/O
- ③ I/O Bus

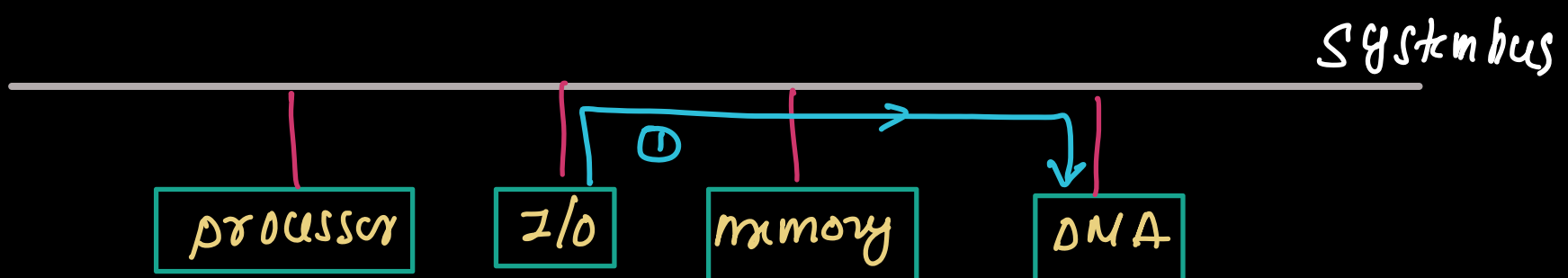
Single bus detached DMA

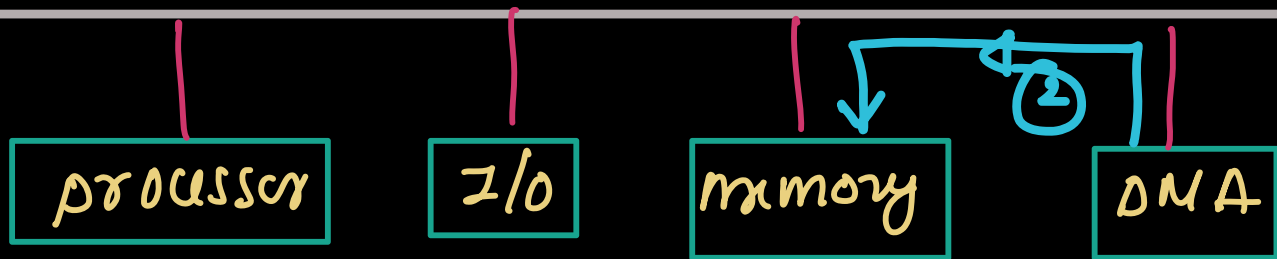
— in this configuration processor, memory, I/O device & DMA are connected to system bus



— Consider an I/O operation, read from I/O device and store in memory.

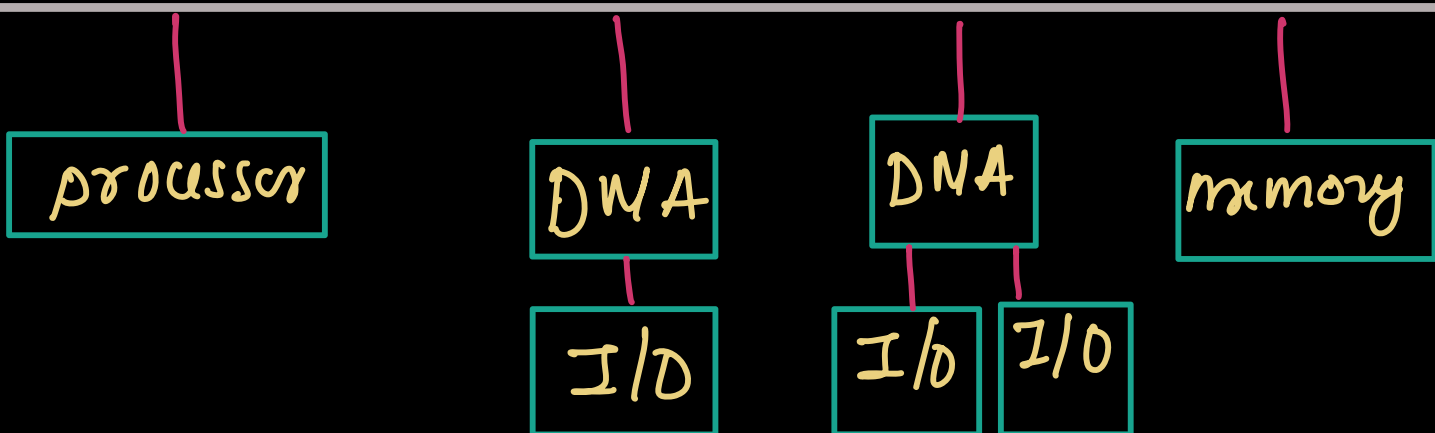
- ① DMA read the data from I/O Module through system Bus
- ② DMA send data to memory through system BUS



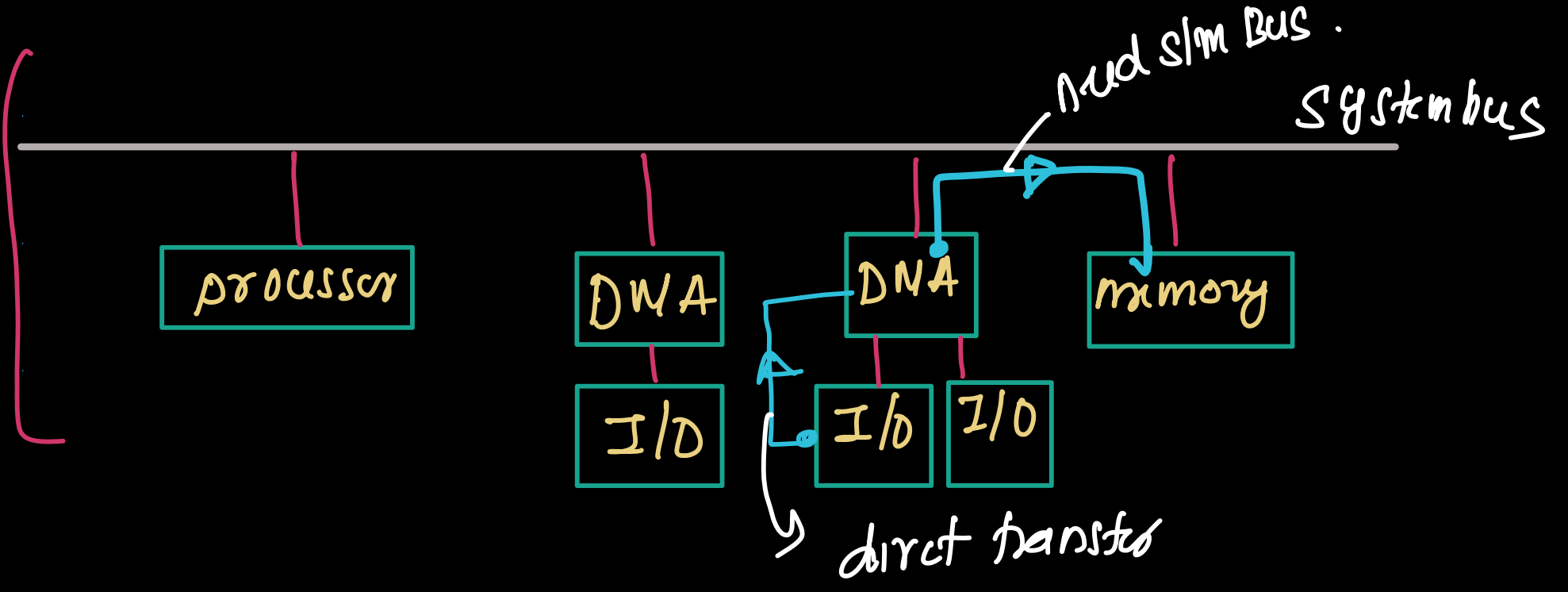


⇒ System bus is used for the both operation, so it's a inefficient configuration.

Single bus with integrated DMA-I/O



- DMA is integrated with one or more I/O (I/O module)
- there is direct connection b/w DMA and I/O module
- Data transfer b/w I/O module and DMA occurs directly without using S/m Bus
- system bus and only when DMA transfer data to memory.



DMA with I/O Bus

- in this configuration, a separate I/O bus is used to connect multiple I/O devices to the DMA.
- processor, memory, DMA share system bus.

